UNIT IV

Fire Prevention & Protection

ignition sources | fire triangle | extinguishing principles | classes of fire | agents | extinguishers | hydrants | alarms | rescue | burns

I. Introduction – what is fire?

Fire is a rapid, self-sustaining **oxidation (combustion)** of a fuel that releases heat and light. Fire is the biggest single threat to plants – it kills people, destroys property and stops production. **Prevention** is cheaper than fire fighting.

Term	Meaning
Flash point	Lowest temperature at which a liquid gives enough vapour to flash (ignite) momentarily when a flame is brought near.
Fire point	Temperature at which vapour continues to burn after ignition (a few degrees above flash point).
Auto-ignition temperature	Temperature at which a substance ignites by itself without any external flame.
Flammable limits (LEL–UEL)	Range of vapour/air concentration that will burn: Lower & Upper Explosive Limits. Eg petrol 1.4–7.6%, LPG 1.8–9.5%, hydrogen 4–75%, acetylene 2.5–81%.
Flammable / Combustible	Flammable liquid: flash point below 65 C (IS/PESO class A < 23 C, B 23–65 C, C 65–93 C).

II. Sources of ignition

Source	Examples
Naked flames	Matches, lighters, burners, welding / cutting torch, furnaces, candles.
Electrical	Short circuit, overloaded cables, loose connections, arcing, faulty motors/heaters, sparks from switches.
Hot surfaces	Boiler, steam pipes, furnace walls, hot bearings, lamps, exhaust manifolds.

Source	Examples
Friction & mechanical sparks	Grinding, hot bearing, tools striking, belt slip, metal-to-metal contact.
Static electricity	Flow of liquids in pipes, filling tanks, conveyor belts, synthetic clothing.
Spontaneous combustion	Oily rags, coal heaps, hay, sawdust, wet grain, linseed oil.
Chemical reaction	Mixing incompatible chemicals, exothermic reactions (oxidisers + fuel), pyrophoric metals.
Lightning & radiant heat	Lightning strike, sun on lenses/glass, radiant heat from nearby fire.
Smoking & arson	Cigarette butts, deliberate fire-raising.
Compression	Rapid compression of gas (diesel engine effect, pneumatic hoses).

Fire prevention measures

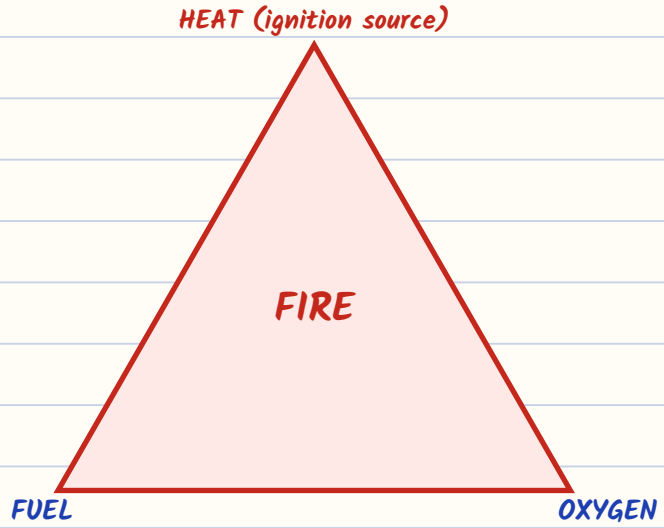
> **Control fuel**: reduce the quantity stored, keep flammables in approved containers/cabinets, away from sources; good housekeeping.

> **Control ignition**: no-smoking rules, hot-work permit, flame-proof (Ex) electrical fittings in hazardous areas, earthing & bonding to remove static, lightning conductors.

> **Control oxygen**: inerting with nitrogen in tanks; closed systems.

> Regular **inspection and maintenance** of electrics and machines; fire safety training, drills and signs.

III. The fire triangle and tetrahedron



Fire triangle = 3 elements

Remove ANY one -> fire dies

Fire tetrahedron adds a 4th:

uninhibited chain reaction

(free radicals) - this is why

halons / DCP stop fire so fast

IV. Principles of fire extinguishing

Remove one side of the fire triangle (or break the chain reaction) and the fire is put out.

Principle	How it works	Agent / method
Starvation	Remove fuel: shut valve, remove combustibles, create firebreak, drain tank	Fuel isolation, fire-break
Smothering	Exclude / dilute oxygen (below ~15%)	Foam blanket, CO2, inert gas, sand, wet blanket
Cooling	Remove heat below ignition temperature	Water jets / sprays, wet chemicals
Chain inhibition	Interrupt free-radical reaction	DCP, halons & halocarbon agents

V. Active and passive fire protection

	Active Fire Protection (AFP)	Passive Fire Protection (PFP)
Meaning	Systems that need action / motion or power to detect and fight fire	Built-in features that contain fire & maintain structure without any action
Examples	Detectors, alarms, sprinklers, hydrants, extinguishers, deluge, gas/foam systems, smoke extractors	Fire-rated walls, floors, doors, fire dampers, fire stops, intumescent coating, fire-proofing of steel, compartmentation, escape routes
Role	Detect, warn, suppress / extinguish	Delay spread, protect structure, give time for escape
Dependence	Needs power, water, human action; can fail	Always ready; no maintenance but periodic inspection

VI. Classes of fire

Class	Fuel involved	Examples	Suitable agent
A	Ordinary combustible solids (leave embers)	Wood, paper, cloth, rubber, plastics	**Water**, foam, ABC DCP
B	Flammable liquids and liquefiable solids	Petrol, oil, paint, grease, solvents, wax	**Foam, DCP, CO2**, halocarbons (not water jet)
C	Flammable gases	LPG, methane, acetylene, hydrogen	Stop gas first; **DCP**, water spray to cool

Class	Fuel involved	Examples	Suitable agent
D	Combustible metals	Magnesium, sodium, potassium, aluminium powder, titanium	**Special dry powder** (graphite, NaCl); never water
E	Electrical equipment (live)	Panels, transformers, motors, wiring	**CO2, DCP**, halocarbon; switch off first; never water/foam

Note

> Some codes use Class **F / K** for cooking oils & fats (wet chemical). Under IS 15683 electrical fire is not a separate class (it is class A/B once power is off) - but your syllabus lists A, B, C, D, E.

VII. Fire extinguishing agents

1. Water

> Cheapest, most common; works by **cooling** (high latent heat) and some smothering by steam (1 litre -> 1700 litres steam).

> Forms: **jet**, **spray / fog**, **sprinkler** (class A). Additives: wetting agents, thickeners, anti-freeze.

> **Not for**: live electrical, oil fires (spreads), metals (Na, K, Mg - releases hydrogen), burning acid.

2. Foam

> Mass of bubbles made by mixing foam concentrate + water + air; floats on liquid to **smother and cool** and suppress vapour. Used on **class B** (and A).

> Types: **Protein (P)**, **Fluoro-protein (FP)**, **AFFF** (aqueous film-forming), **FFFP**, **AR-AFFF** (alcohol-resistant) for polar solvents, **High-expansion** foam for enclosed spaces.

> Expansion ratio: low (up to 20:1), medium (20-200:1), high (200-1000:1). Mix: 3% or 6% induction.

> **Not for** live electrical, gas fires or 3-D pressure fires.

3. Dry chemical powder (DCP)

> Fine powder expelled by gas; **inhibits chain reaction** and smothers. Very fast knock-down of flames.

> **Sodium bicarbonate** (BC), **Potassium bicarbonate / Purple-K** (BC, high efficiency), **Monoammonium phosphate** (ABC, multi-purpose), Monnex (urea + K bicarbonate).

> Non-conductive; ok for electrical. Cons: no cooling (re-ignition), messy, corrosive; ABC powder damages electronics & is poor in confined space.

4. Carbon dioxide (CO2)

> Colourless, odourless, non-conductive gas stored as liquid at 50-60 bar; expands to 450 times volume; **smothers** (reduces oxygen) and cools (snow).

> Used for **class B and E** - electrical rooms, computer rooms, engines, flammable liquids. No residue.

> **Hazard**: asphyxiation in closed places above 9-10% (alarms, pre-discharge delay); not for class A deep-seated or metals (Mg burns in CO2).

5. Halons, halon alternatives, inert gases

> **Halons** (Halon 1211, 1301, 2402): hydrocarbons with bromine; best chain-breakers, clean, safe on electronics - but **ozone-depleting**, banned by **Montreal Protocol (1987)**; production stopped in 1994 (developed countries), phase-out in India by 2010.

Alternative	Examples	Features
Halocarbon (HFC)	HFC-227ea (FM-200), HFC-125 (FE-25), HFC-23, HFC-236fa	Chain inhibition + cooling; zero ozone depletion but some global-warming potential; 6-10% design conc.; electrical rooms, data centres.
Fluoroketone	FK-5-1-12 (Novec 1230)	Liquid at room temp; GWP = 1; low toxicity; 4-6% design concentration.
Inert gases	IG-541 (Inergen: N2 52%, Ar 40%, CO2 8%), IG-55 (N2+Ar 50:50), IG-100 (N2), IG-01 (Ar)	Reduce oxygen to ~12-14%; natural, no residue, no ozone/GWP; needs many large cylinders and pressure venting.
Water mist	High-pressure micro-droplets	Cooling + oxygen displacement; minimal water damage.

6. Dry powders for metal fires (class D)

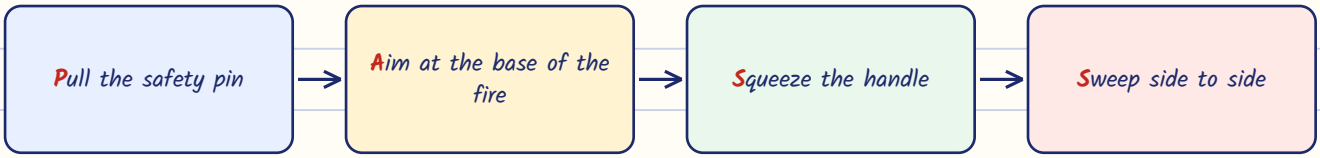
> Graphite, sodium chloride (Met-L-X), copper powder, soda ash, dry sand/dolomite - smother and form crust. Apply gently and cover; do not use water, CO2 or halogenated agents.

VIII. Types of fire extinguishers

A **portable fire extinguisher** is a cylinder containing an agent and expelling propellant; operated by hand to put out small fires. A fire must be attacked within the first **3-4 minutes**.

Type	Agent / colour band (EN 3)	Use on	Not for
Water (gas cartridge / stored pressure)	Water; red	A	B, C, D, E
Foam (AFFF)	Foam; cream	A, B	E, C, D
DCP (ABC / BC)	Powder; blue	A,B,C,E	D; confined space
CO2	Carbon dioxide; black	B, E (small A)	D, deep-seated A
Wet chemical	Potassium acetate; yellow	F (cooking oil), A	E
Clean agent (HFC)	Halocarbon; green	B, E, A	D

Operation of an extinguisher – PASS



- Stand 2-3 m upwind with escape route behind you; fires should be **small**; raise alarm first.
- Maintenance: monthly check (seal, gauge, nozzle), yearly service, hydro-test every 5 years, refill after use; mounted at 1 m height near exit, 1 per 200-250 m2 (travel distance 15-30 m).
- Fixed systems: **automatic sprinkler**, deluge, foam, CO2 total-flooding.

Fire stoppers (fire stops / firestopping)

Fire stops are **passive barriers** that seal gaps and openings around pipes, cables, ducts and joints in fire-rated walls and floors so that **flame, smoke and hot gases** cannot spread from one compartment to another.

- Materials: intumescent sealants and collars (swell on heat), mineral-wool + sealant, fire-resistant mortar, fire pillows, cable-transit blocks.
- **Fire dampers** in ventilation ducts close automatically (fusible link) when temperature reaches about 70 C.
- **Fire doors** (self-closing, rated 1-2 hr), fire-rated shutters, cavity barriers in roof/ceiling voids.

IX. Hydrant system, hoses, monitors and stand pipes

Hydrant pipes

> Underground / above-ground **ring main** (150-300 mm dia, steel/ CI) around the plant with sectional valves; fed by fire pumps (electric + diesel standby) and storage tank (static water tank).

> **Hydrants** (yard or landing valve type): pillar hydrants spaced about 30-45 m, with 63 mm outlets; each with hose box (2 hoses of 15 m, branch pipe, key).

> Pressure at the farthest hydrant: about **3.5 kg/cm2 (3.5 bar)**, flow of 2850 lpm for a main pump.

> Pipes painted **red**; tested and flushed periodically; valves exercised.

Hoses

Type	Details
Delivery hose	63 mm (2.5 in) - 15 m lengths, instantaneous couplings; **rubber-lined (RRL), canvas, PVC / polyurethane lay-flat**; type A,B,C by IS 636.
Hose reel	20-25 mm semi-rigid, 36 m, small fires, first-aid fire fighting by trained persons.
Suction hose	Hard rubber-lined, steel-wire reinforced to draw water from tank/pond.
Care	Dry after use, no kinks, roll properly, test at 15 kg/cm2 yearly, store in hose box.

Monitors

> **Fire monitor (water cannon)**: fixed or trailer-mounted nozzle with large throw (30-60 m), 1000-4000 lpm; manual / remote / oscillating; water or **foam**. Used on tank farms, refineries, loading bays.

> Placed at a safe distance so that crew need not go close to heat.

Fire watchers

> A **fire watcher** is a trained person assigned to watch for fire during **hot work** (welding, cutting, grinding) or high-risk periods.

> Duties: remove combustibles within 10 m, keep extinguisher ready, watch sparks and adjacent areas, raise alarm, and watch for **at least 30 minutes after** work ends (smouldering).

Layout of stand pipes (risers)

A **standpipe** is a vertical water pipe in a multi-storey building with outlets on every floor for fire fighting.

Type	Details
Dry riser	Empty pipe, charged by fire brigade pumps through inlet at ground floor; for buildings 15-24 m tall; 100 mm (up to 30 m) or 150 mm.
Wet riser	Permanently full of water, connected to pumps and tank; **for tall buildings > 24 m**; each floor has landing valve (63 mm) in the staircase lobby.

Type	Details
Down-comer	Wet pipe running down from the terrace tank; supplies hose on each floor by gravity; first-aid purpose.
Layout	Risers in staircase enclosures/ lobbies, one per 1000 m2 of floor area; hose reel, landing valve, 2 hoses and branch pipe; test point at terrace; combined yard hydrant + fire brigade connection.

X. Fire station, alarms and sirens

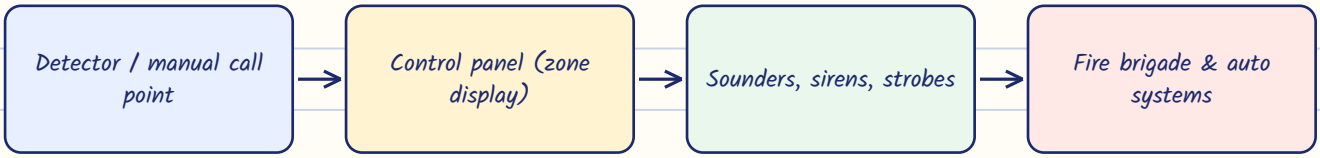
Fire station (plant fire station)

> **Location**: near the main gate/ road, central to high-risk areas, with clear access; response within 3-5 min to any point.

> Facilities: fire tender bay, water tender, foam tender, rescue van, control room (24 hrs), call receiving system, hose-drying tower, workshop, drill yard, living quarters for crew, breathing-apparatus room.

> Crew: fire officer, leaders, drivers, firemen on shift; link to the **city fire brigade** (mutual aid).

Fire alarm systems



Detector	Principle	Best for
Heat (fixed-temp / rate-of-rise)	Senses temperature (57-90 C) or fast rise; fusible element, bimetal	Kitchens, boiler rooms, dusty areas
Smoke - ionisation	Smoke reduces ion current	Fast flaming fires
Smoke - optical	Smoke scatters / obscures light beam	Smouldering fires, offices
Flame (UV / IR)	Detects radiation from flame in ms	Fuel handling, tank farms
Gas / VAD	Detects combustible gas or air sample	Process plants, data rooms

> **Manual call points** (break-glass) near exits, every 30 m; **sirens/hooters** audible 85 dB (or 65 dB above ambient) with distinct tone; **public address** and strobe lights for deaf.

> Zones, addressable systems, battery back-up, monthly test.

XI. Maintenance of fire trucks (fire tenders)

A fire tender must be **ready to move in less than 1 minute**. Preventive maintenance and daily checks are therefore compulsory.

Frequency	Checks
Daily / shift	Fuel, engine oil, coolant, battery, tyre pressure, brakes, lights & siren; water tank full; foam tank level; hoses, branch pipes, ladders, extinguishers, breathing apparatus in place.
Weekly	Start & run pump (priming, pressure, relief valve), PTO (power take-off) engagement, vacuum test, hose-reel, monitor operation, grease points.
Monthly	Test foam inductor & proportioner, check hose condition, valves leak-free, battery charge, cleaning of strainers.
Yearly	Pump performance & pressure test, hydro-test hoses and ladders, service by workshop, tyre rotation, painting & reflective marking.

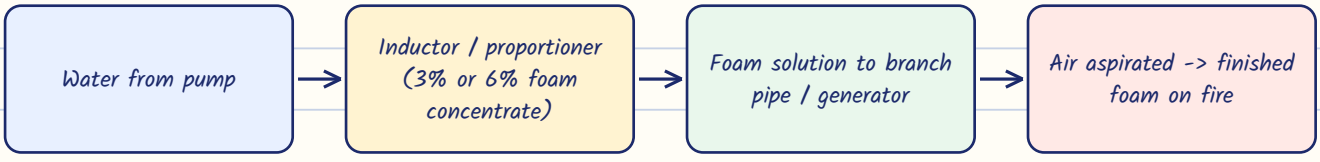
> After every use: wash and dry hoses, refill water/ foam / fuel, replace used equipment, record in the log-book.

> Records: log-book of fuel, running hours, defects, test results; trained drivers (driver-pump operators) with regular drills.

> Fire engine types: **water tender, foam tender, dry-powder tender, rescue tender, turn-table ladder, hydraulic platform, ambulance**.

XII. Foam generators

A **foam generator** (foam maker) produces foam from **foam concentrate + water + air**; the foam is then thrown on the burning liquid.



Type	Working	Use
Foam branch pipe (self-aspirating)	Water + concentrate through branch; air drawn in by venturi effect	Low-expansion foam for spill fires

Type	Working	Use
Foam generator / pourer on tank	Foam is delivered to the tank wall or by a chamber (fixed on tank top)	Fixed protection of storage tanks
High-expansion generator	A fan blows air through a screen sprayed with solution; expansion up to 1000:1	Warehouses, basements, cable tunnels, ship holds
Balanced pressure proportioner	Pump supplies concentrate in exact proportion to water flow	Refineries, fixed systems

> Foam application rates: about **4-6 L/min/m2** on hydrocarbon spill; **subsurface injection** for large tank fires; check expiry of concentrate (shelf life 8-10 yrs).

> Foam should be applied gently on the surface, from upwind, not directed into liquid.

XIII. Escape from fire & rescue operations

Escape from fire - golden rules

1. **Raise the alarm** - shout 'Fire!', operate call point, inform fire station / call 101.
2. **Leave immediately**; do not stop for belongings; help the disabled.
3. **Feel closed doors** with the back of the hand; if hot, do not open; use another route.
4. Close doors behind you to slow the fire and smoke.
5. Smoke rises: **keep low**, crawl, cover nose & mouth with wet cloth.
6. Use **stairs, never lifts**; follow green exit signs to the assembly point and report to the warden.
7. If trapped: go to room with window, seal gaps with wet cloth, signal for help. **Stop, Drop and Roll** if clothes catch fire.
8. Never re-enter a burning building.

Rescue operations

> Rescue crew wear **breathing apparatus (BA), fire-proximity suits, helmets, torch, life-line**; work in pairs, with a control point and entry-board.

> Priority order: **life -> property -> environment**. Search systematically (right-hand wall rule), mark searched rooms.

> Carry methods: drag, fireman's lift, two-hand seat, chair carry; use of ladders, ropes, escape chutes, jumping sheets, hydraulic platforms.

> Give first aid and oxygen at once and hand over to ambulance; keep a casualty record and do a head-count.

XIV. Fire drills

Fire drill = rehearsal of the emergency plan so that everyone knows what to do and how to leave in a short time. Done at least **twice a year** (once unannounced) and after any change of layout.

Element	Detail
Fire warden / marshal	Responsible persons for each area & floor; check rooms, guide staff, close doors, report to chief warden.
Assembly point	Safe open place away from the building; head-count by roll-call.
Evacuation	Alarm, shut down plant & isolate energy (if safe), leave by marked routes, disabled assisted.
Fire fighting	Fire team with extinguishers / hose reels; fire brigade informed and guided.
Review	Time taken recorded (target: building clear in 3 min), problems noted, corrections made, training updated.

Emergency organisation

> **Chief controller / Site main controller** -> Incident controller -> Fire team, rescue team, first-aid team, security, communications, evacuation & head-count team.

> On-site emergency plan is mandatory for hazardous process units (Factories Act Sec 41B) and off-site plan by the district authorities.

XV. First aid for burns

Degree	Depth / appearance	Example
1st degree	Epidermis only; red, painful, dry, no blisters	Sunburn, brief flame contact
2nd degree	Dermis; blisters, wet, very painful	Hot liquid scald
3rd degree	Full thickness; white / charred, leathery, painless (nerves destroyed)	Prolonged flame, electrical burn

Treatment - 'COOL, COVER, CALL'

1. **Stop the burning**: remove from heat; **Stop-Drop-Roll** / smother flames with blanket (not nylon).
2. **Cool**: hold under clean **running cool water for 10-20 minutes** (not ice, not very cold).
3. Remove rings, watch, belt and loose clothing **before swelling**; do not pull off clothes stuck to the

burn.

4. **Cover**: loosely with a clean, **non-fluffy, non-stick dressing** / cling film / clean cloth.
5. Treat shock: lay down, raise legs, warm; give sips of water if conscious and no vomiting.
6. **Call** medical help / ambulance (108) for large, deep burns, face, hands, genital, chemical, electrical burns, children.

DON'T

> Do not apply butter, oil, toothpaste or ointment; do not use ice; do not burst blisters; do not touch the burn with bare hands.

> **Chemical burns**: flush with plenty of water for 20 min or more (remove contaminated clothes); **electrical burns**: switch OFF power first and check heart.

Burn > 10% body area (adult) or 3rd degree = hospital

XVI. Quick revision & important questions

Remember

> Fire triangle: heat + fuel + oxygen (+ chain reaction = tetrahedron).

> Classes: A solids, B liquids, C gases, D metals, E electrical.

> Extinguishing: starvation, smothering, cooling, chain inhibition.

> Halons banned (Montreal Protocol 1987); alternatives: HFC-227ea, Novec 1230, Inergen.

> PASS: Pull, Aim, Squeeze, Sweep. Wet riser for buildings > 24 m.

> Burns: cool under running water 10-20 min, cover, call.

1. Explain the fire triangle and the principles of fire extinguishing. (10 M)
2. Classify fires and name the extinguishing agent suitable for each. (10 M)
3. Discuss water, foam, DCP and CO2 as fire extinguishing agents. (10 M)
4. What are Halon alternatives? Explain halocarbons and inert gases. (5 M)
5. Describe the types of portable fire extinguishers and their operation. (10 M)
6. Explain the hydrant system, hoses, monitors and stand pipes. (10 M)
7. Write notes on fire alarm, fire station, fire drills, rescue and first aid for burns. (5 M each)